

and write F for the set of its edges. If $F = E(G)$, then W is an Euler tour. Suppose, therefore, that $G' := G - F$ has an edge.

For every vertex $v \in G$, an even number of the edges of G at v lies in F , so the degrees of G' are again all even. Since G is connected, G' has an edge e incident with a vertex on W . By the induction hypothesis, the component C of G' containing e has an Euler tour. Concatenating this with W (suitably re-indexed), we obtain a closed walk in G that contradicts the maximal length of W . $\square$

1.9 Some linear algebra

[8.7]

Let $G = (V, E)$ be a graph with n vertices and m edges, say $V = \{v_1, \dots, v_n\}$ and $E = \{e_1, \dots, e_m\}$. The *vertex space* $\mathcal{V}(G)$ of G is the vector space over the 2-element field $\mathbb{F}_2 = \{0, 1\}$ of all functions $V \rightarrow \mathbb{F}_2$. Every element of $\mathcal{V}(G)$ corresponds naturally to a subset of V , the set of those vertices to which it assigns a 1, and every subset of V is uniquely represented in $\mathcal{V}(G)$ by its indicator function. We may thus think of $\mathcal{V}(G)$ as the power set of V made into a vector space: the sum $U + U'$ of two vertex sets $U, U' \subseteq V$ is their symmetric difference (why?), and $U = -U$ for all $U \subseteq V$. The zero in $\mathcal{V}(G)$, viewed in this way, is the empty (vertex) set $\emptyset$. Since $\{\{v_1\}, \dots, \{v_n\}\}$ is a basis of $\mathcal{V}(G)$, its *standard basis*, we have $\dim \mathcal{V}(G) = n$.

 $G = (V, E)$ vertex
space $\mathcal{V}(G)$

+

In the same way as above, the functions $E \rightarrow \mathbb{F}_2$ form the *edge space* $\mathcal{E}(G)$ of G : its elements correspond to the subsets of E , vector addition amounts to symmetric difference, $\emptyset \subseteq E$ is the zero, and $F = -F$ for all $F \subseteq E$. As before, $\{\{e_1\}, \dots, \{e_m\}\}$ is the *standard basis* of $\mathcal{E}(G)$, and $\dim \mathcal{E}(G) = m$. Given two elements F, F' of the edge space, viewed as functions $E \rightarrow \mathbb{F}_2$, we write

edge space
 $\mathcal{E}(G)$ standard
basis

$$\langle F, F' \rangle := \sum_{e \in E} F(e)F'(e) \in \mathbb{F}_2.$$

 $\langle F, F' \rangle$

This is zero if and only if F and F' have an even number of edges in common; in particular, we can have $\langle F, F \rangle = 0$ with $F \neq \emptyset$. Given a subspace $\mathcal{F}$ of $\mathcal{E}(G)$, we write

$$\mathcal{F}^\perp := \{ D \in \mathcal{E}(G) \mid \langle F, D \rangle = 0 \text{ for all } F \in \mathcal{F} \}.$$

 $\mathcal{F}^\perp$

This is again a subspace of $\mathcal{E}(G)$ (the space of all vectors solving a certain set of linear equations – which?), and one can show that

$$\dim \mathcal{F} + \dim \mathcal{F}^\perp = m.$$

cycle space
 $\mathcal{C}(G)$

The *cycle space* $\mathcal{C} = \mathcal{C}(G)$ is the subspace of $\mathcal{E}(G)$ spanned by all the cycles in G – more precisely, by their edge sets.¹⁰ The dimension of $\mathcal{C}(G)$ is sometimes called the *cyclomatic number* of G .

The elements of $\mathcal{C}$ are easily recognized by the degrees of the subgraphs they form. Moreover, to generate the cycle space from cycles we only need disjoint unions rather than arbitrary symmetric differences:

[4.5.1]
[8.7.3]

Proposition 1.9.1. *The following assertions are equivalent for edge sets $D \subseteq E$:*

- (i) $D \in \mathcal{C}(G)$;
- (ii) D is a (possibly empty) disjoint union of edge sets of cycles in G ;
- (iii) All vertex degrees of the graph (V, D) are even.

Proof. Since cycles have even degrees and taking symmetric differences preserves this, (i)→(iii) follows by induction on the number of cycles used to generate D . The implication (iii)→(ii) follows by induction on $|D|$: if $D \neq \emptyset$ then (V, D) contains a cycle C , whose edges we delete for the induction step. The implication (ii)→(i) is immediate from the definition of $\mathcal{C}(G)$. $\square$

cut, sides
cross
bond
atomic

A set F of edges is a *cut* in G if there exists a partition¹¹ $\{V_1, V_2\}$ of V such that $F = E(V_1, V_2)$. If G is connected, this partition is unique given F (Exercise 45), and the edges in F are said to *cross* it. Its classes V_1, V_2 are the *sides* of the cut F . A minimal non-empty cut is a *bond*. Cuts or bonds of the form $E(v)$ are *atomic*.

[4.6.3]

Proposition 1.9.2. *Together with $\emptyset$, the cuts in G form a subspace $\mathcal{B} = \mathcal{B}(G)$ of $\mathcal{E}(G)$. This space is generated by atomic cuts.*

Proof. Let $\mathcal{B}$ denote the subspace of $\mathcal{E}(G)$ generated by its atomic cuts. Every cut of G , with vertex partition $\{V_1, V_2\}$ say, equals $\sum_{v \in V_1} E(v)$ and hence lies in $\mathcal{B}$. Conversely, every set $\sum_{v \in U} E(v) \in \mathcal{B}$ is either empty, e.g. if $U \in \{\emptyset, V\}$, or it is the cut $E(U, V \setminus U)$. $\square$

cut space
 $\mathcal{B}(G)$

The space $\mathcal{B}$ from Proposition 1.9.2 is the *cut space*, or *bond space*, of G . It is not difficult to find among the atomic cuts an explicit basis for $\mathcal{B}$, and thus to determine its dimension (Exercise 48). Note that the bonds are for $\mathcal{B}$ what cycles are for $\mathcal{C}$: the minimal non-empty elements.

The ‘non-empty’ condition in the definition of a bond bites only if G is disconnected. If G is connected, its bonds are just its minimal cuts,

¹⁰ For simplicity, we shall not always distinguish between the edge sets $F \in \mathcal{E}(G)$ and the subgraphs (V, F) they induce in G . When we wish to be more precise, such as in Chapter 8.6, we shall use the word ‘*circuit*’ for the edge set of a cycle.

¹¹ Recall that partition classes in this book are non-empty. The empty set of edges, therefore, is a cut only if the graph is disconnected.

and these are easy to recognize: a cut in a connected graph is minimal if and only if both sides of the corresponding vertex partition induce connected subgraphs (Exercise 41). If G is disconnected, its bonds are the minimal cuts of its components.

In analogy to Proposition 1.9.1, bonds and disjoint unions suffice to generate the cut space:

Lemma 1.9.3. *Every cut is a disjoint union of bonds.*

[4.6.2]
[6.5.2]

Proof. We apply induction on the size of the cut F considered. For $F = \emptyset$ the assertion is trivial (with the empty union). If $F \neq \emptyset$ is not itself a bond, it properly contains some other non-empty cut F' . By Proposition 1.9.2, also $F \setminus F' = F + F'$ is a smaller non-empty cut. By the induction hypothesis, both F' and $F \setminus F'$ are disjoint unions of bonds, and hence so is F . $\square$

Exercise 47 indicates how to construct the bonds for Lemma 1.9.3 explicitly. In Chapter 3.1 we shall prove some more details about the possible positions of the cycles and bonds of a graph within its overall structure (Lemmas 3.1.2 and 3.1.3).

Theorem 1.9.4. *The cycle space $\mathcal{C}$ and the cut space $\mathcal{B}$ of any graph satisfy*

[4.6]

$$\mathcal{C} = \mathcal{B}^\perp \quad \text{and} \quad \mathcal{B} = \mathcal{C}^\perp.$$

Proof. Consider a graph $G = (V, E)$. Clearly, any cycle in G has an even number of edges in each cut. This implies $\mathcal{C} \subseteq \mathcal{B}^\perp$ and $\mathcal{B} \subseteq \mathcal{C}^\perp$.

(1.6.1)
(1.10)

To prove $\mathcal{B}^\perp \subseteq \mathcal{C}$, recall from Proposition 1.9.1 that for every edge set $F \notin \mathcal{C}$ there exists a vertex v incident with an odd number of edges in F . Then $\langle E(v), F \rangle = 1$, so $E(v) \in \mathcal{B}$ implies $F \notin \mathcal{B}^\perp$. This completes the proof of $\mathcal{C} = \mathcal{B}^\perp$.

To prove $\mathcal{C}^\perp \subseteq \mathcal{B}$, let $F \in \mathcal{C}^\perp$ be given. Consider the multigraph¹² H obtained from G by contracting the edges in $E \setminus F$. Any cycle in H has all its edges in F . Since we can extend it to a cycle in G by edges from $E \setminus F$, the number of these edges must be even. Hence H is bipartite, by Proposition 1.6.1. Its bipartition induces a bipartition (V_1, V_2) of V such that $E(V_1, V_2) = F$, showing $F \in \mathcal{B}$ as desired. $\square$

Consider a connected graph $G = (V, E)$ with a spanning tree $T \subseteq G$. For every chord $e \in E \setminus E(T)$ there is a unique cycle C_e in $T + e$, the *fundamental cycle* of e with respect to T . Similarly, for every edge $f \in T$ the forest $T - f$ has exactly two components (Theorem 1.5.1 (iii)). The

fundamental
cycle/cut

(1.5.1)

¹² See Section 1.10: such contractions might create loops in F , but bipartite multigraphs have no loops. The proof of Proposition 1.6.1 works for multigraphs too.

set $D_f \subseteq E$ of edges of G between these components is a bond in G , the *fundamental cut* of f with respect to T .

Notice that $f \in C_e$ if and only if $e \in D_f$, for all edges $e \notin T$ and $f \in T$. This is an indication of some deeper duality, which the following theorem explores further.

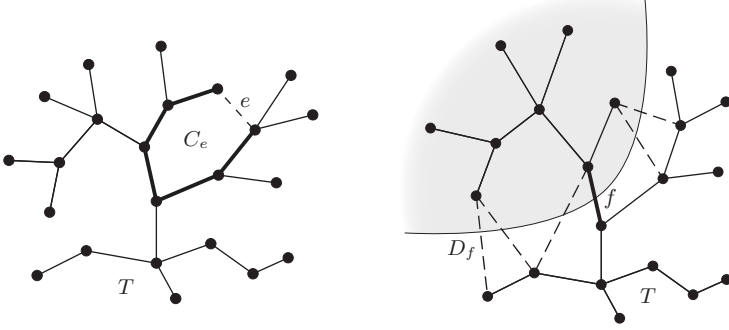


Fig. 1.9.1. The fundamental cycle C_e , and the fundamental cut D_f

[4.5.1] **Theorem 1.9.5.** Let G be a connected graph with n vertices and m edges, and let $T \subseteq G$ a spanning tree.

- (i) The fundamental cuts and cycles of G with respect to T form bases of $\mathcal{B}(G)$ and $\mathcal{C}(G)$, respectively.
- (ii) Hence, $\dim \mathcal{B}(G) = n - 1$ and $\dim \mathcal{C}(G) = m - n + 1$.

(1.5.2) *Proof.* (i) Note that an edge $f \in T$ lies in D_f but in no other fundamental cut, while an edge $e \notin T$ lies in C_e but in no other fundamental cycle. Hence the fundamental cuts and cycles form linearly independent sets in $\mathcal{B} = \mathcal{B}(G)$ and $\mathcal{C} = \mathcal{C}(G)$, respectively.

Let us show that the fundamental cycles generate every cycle C . By our initial observation, $D := C + \sum_{e \in C \setminus T} C_e$ is an element of $\mathcal{C}$ that contains no edge outside T . But by Proposition 1.9.1, the only element of $\mathcal{C}$ contained in T is $\emptyset$. So $D = \emptyset$, giving $C = \sum_{e \in C \setminus T} C_e$.

Similarly, every cut D is a sum of fundamental cuts. Indeed, the element $D + \sum_{f \in D \cap T} D_f$ of $\mathcal{B}$ contains no edge of T . As $\emptyset$ is the only element of $\mathcal{B}$ missing T , this implies $D = \sum_{f \in D \cap T} D_f$.

(ii) By (i), the fundamental cuts and cycles form bases of $\mathcal{B}$ and $\mathcal{C}$. As there are $n - 1$ fundamental cuts (Corollary 1.5.2), there are $m - n + 1$ fundamental cycles. $\square$

incidence
matrix

The *incidence matrix* $B = (b_{ij})_{n \times m}$ of a graph $G = (V, E)$ with $V = \{v_1, \dots, v_n\}$ and $E = \{e_1, \dots, e_m\}$ is defined over $\mathbb{F}_2$ by

$$b_{ij} := \begin{cases} 1 & \text{if } v_i \in e_j \\ 0 & \text{otherwise.} \end{cases}$$

As usual, let $B^\top$ denote the transpose of B . Then B and $B^\top$ define linear maps $B: \mathcal{E}(G) \rightarrow \mathcal{V}(G)$ and $B^\top: \mathcal{V}(G) \rightarrow \mathcal{E}(G)$ with respect to the standard bases. As is easy to check, B maps an edge set $F \subseteq E$ to the set of vertices incident with an odd number of edges in F , while $B^\top$ maps a set $U \subseteq V$ to set of edges with exactly one end in U . In particular:

Proposition 1.9.6.

- (i) The kernel of B is $\mathcal{C}(G)$.
- (ii) The image of $B^\top$ is $\mathcal{B}(G)$. □

More on this in the exercises and notes at the end of this chapter.

The *adjacency matrix* $A = (a_{ij})_{n \times n}$ of G is defined by

*adjacency
matrix*

$$a_{ij} := \begin{cases} 1 & \text{if } v_i v_j \in E \\ 0 & \text{otherwise.} \end{cases}$$

Viewed as a linear map $\mathcal{V} \rightarrow \mathcal{V}$, the adjacency matrix maps a given set $U \subseteq V$ to the set of vertices with an odd number of neighbours in U .

Let D denote the real diagonal matrix $(d_{ij})_{n \times n}$ with $d_{ii} = d(v_i)$ and $d_{ij} = 0$ otherwise. Our last proposition establishes a connection between A and B (now viewed as real matrices), which can be verified simply from the definition of matrix multiplication:

Proposition 1.9.7. $BB^\top = A + D$. □

It is also instructive to check that $A + D$, with entries taken mod 2, defines the same map $\mathcal{V} \rightarrow \mathcal{V}$ as the composition of the maps of B and $B^\top$ (Exercise 57).

1.10 Other notions of graphs

For completeness, we now mention a few other notions of graphs which feature less frequently or not at all in this book.

A *hypergraph* is a pair (V, E) of disjoint sets, where the elements of E are non-empty subsets (of any cardinality) of V . Thus, graphs are special hypergraphs.

hypergraph

A *directed graph* (or *digraph*) is a pair (V, E) of disjoint sets (of *vertices* and *edges*) together with two maps $\text{init}: E \rightarrow V$ and $\text{ter}: E \rightarrow V$ assigning to every edge e an *initial vertex* $\text{init}(e)$ and a *terminal vertex* $\text{ter}(e)$. The edge e is said to be *directed from* $\text{init}(e)$ *to* $\text{ter}(e)$. Note that a directed graph may have several edges between the same two vertices x, y . Such edges are called *multiple edges*; if they have the same direction (say from x to y), they are *parallel*. If $\text{init}(e) = \text{ter}(e)$, the edge e is called a *loop*.

*directed
graph*

$\text{init}(e)$
 $\text{ter}(e)$

loop